P2P Process Analysis
SLA, Bottlenecks & Risk

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Executive Summary - P2P Process Delay Analysis

Key Findings

- SLA performance on closed cases remains relatively stable; however, the share of open cases **increased materially in recent periods**, indicating growing execution backlog.
- The main structural **bottleneck is** the Invoice Receipt → Clear Invoice stage, which contributes the largest share of total lead time in delayed cases.
- **Delay risk** is concentrated across selected segments: top vendors and top spend areas (packaging) account for the majority of delayed cases.
- One company **entity represents nearly all delayed volume**, suggesting local process or governance issues.

Recommended Actions

- Prioritise reduction of backlog in ageing open cases (>90 days).
- **Review** invoice clearing workflow and approval cycle in the IR → CI stage.
- Launch targeted deep-dive for Company 4, top vendors, and Packaging-related spend areas.

KPI Snapshot

- Analysed cases: 139,549
- **Delay threshold: >75 days**
- Main bottleneck: IR → CI
- **Top 15 vendors: 52% of delayed cases**
- **Top 10 spend areas: 82% of delayed cases**

Is SLA performance deteriorating?

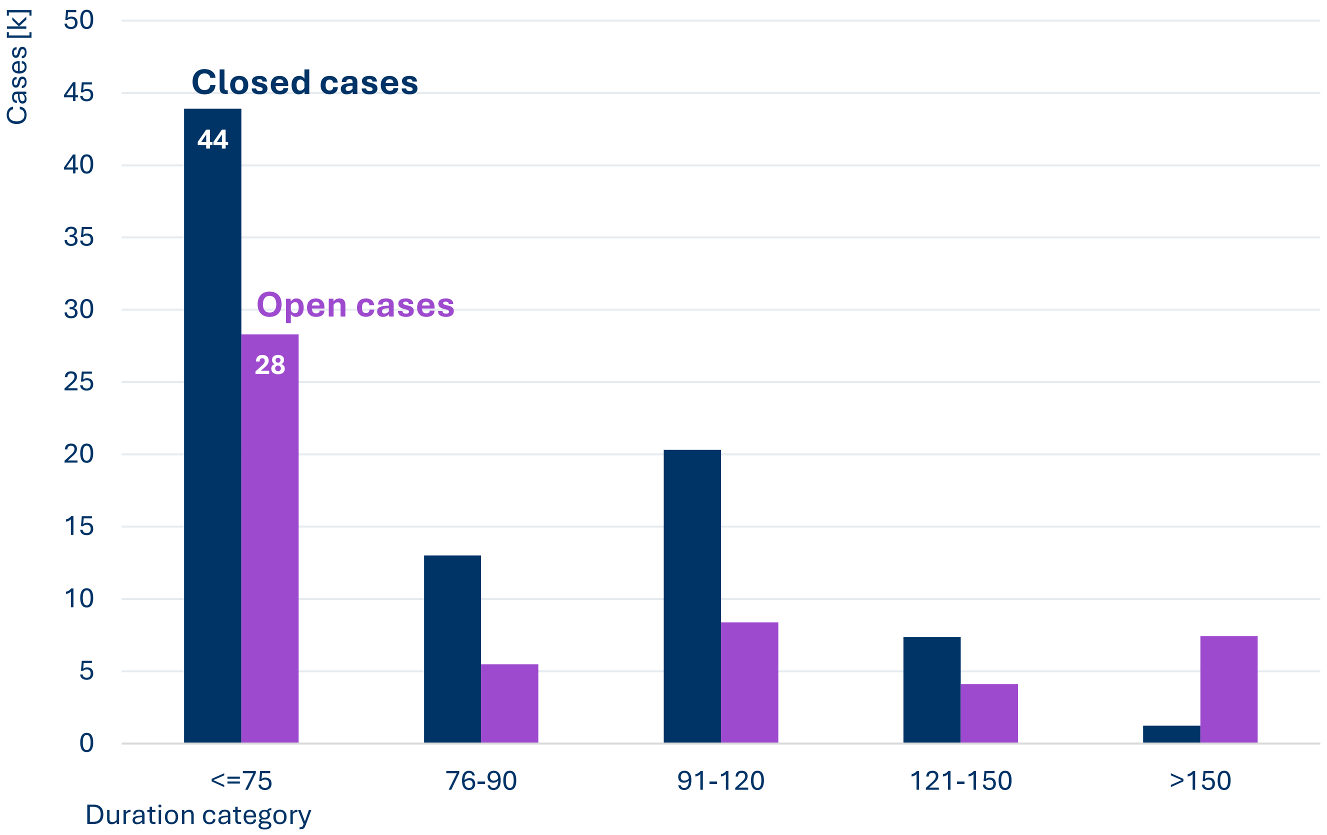
Lead time vs backlog risk

Snapshot status for Jan-2019 + 2H 2018.

Status Overview

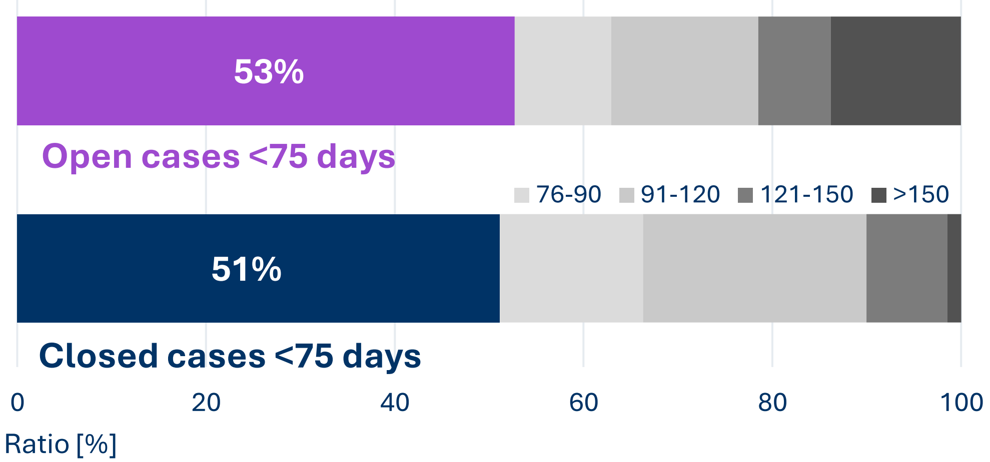
SLA Performance vs Backlog Evaluation

Most open cases are still within the SLA window but require active monitoring to avoid breaching the limit.



Ratio distribution per duration category

In last 6 months **over 50% cases was closed within the SLA window.**



Duration category	Open Cases		Closed Cases	
	Total Number	Ratio [%]	Total Number	Ratio [%]
<=75	28,309	52.71	43,900	51.14
76-90	5,486	10.22	13,021	15.17
91-120	8,374	15.59	20,304	23.65
121-150	4,111	7.65	7,377	8.59
>150	7,424	13.82	1,243	1.45
Total	53,704	100.00	85,845	100.00

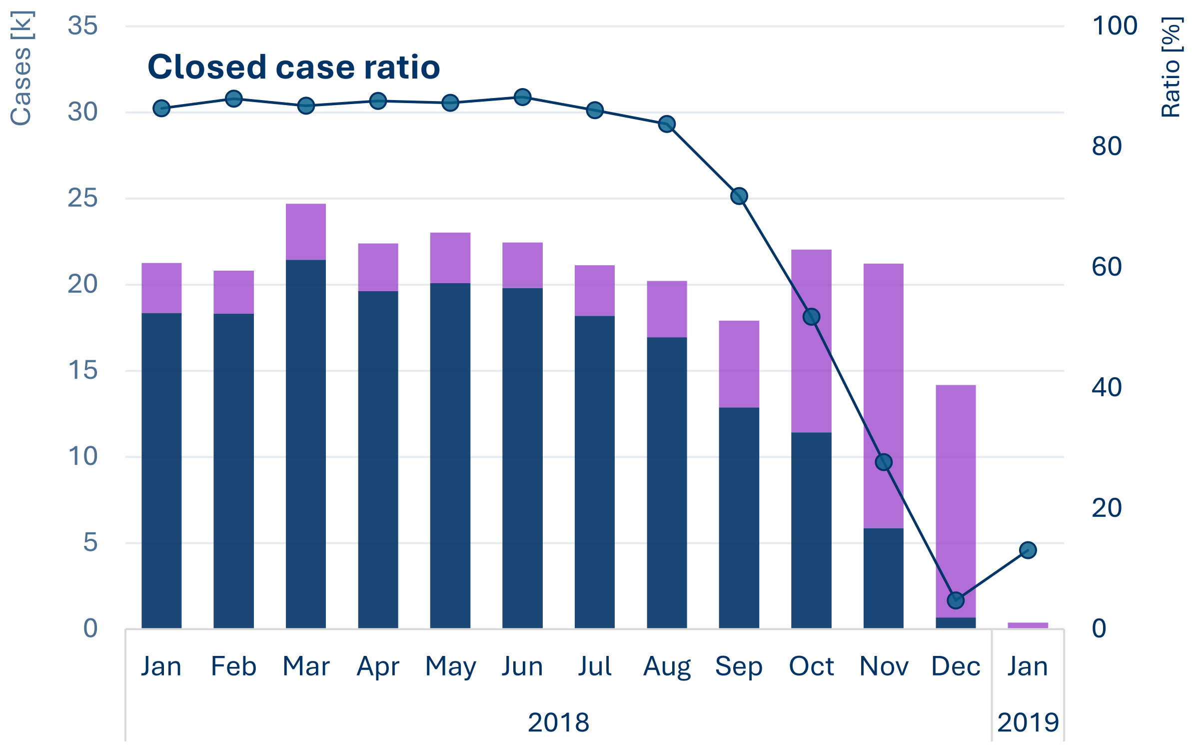
Is SLA performance deteriorating?

Lead time vs backlog risk

Monthly Trend

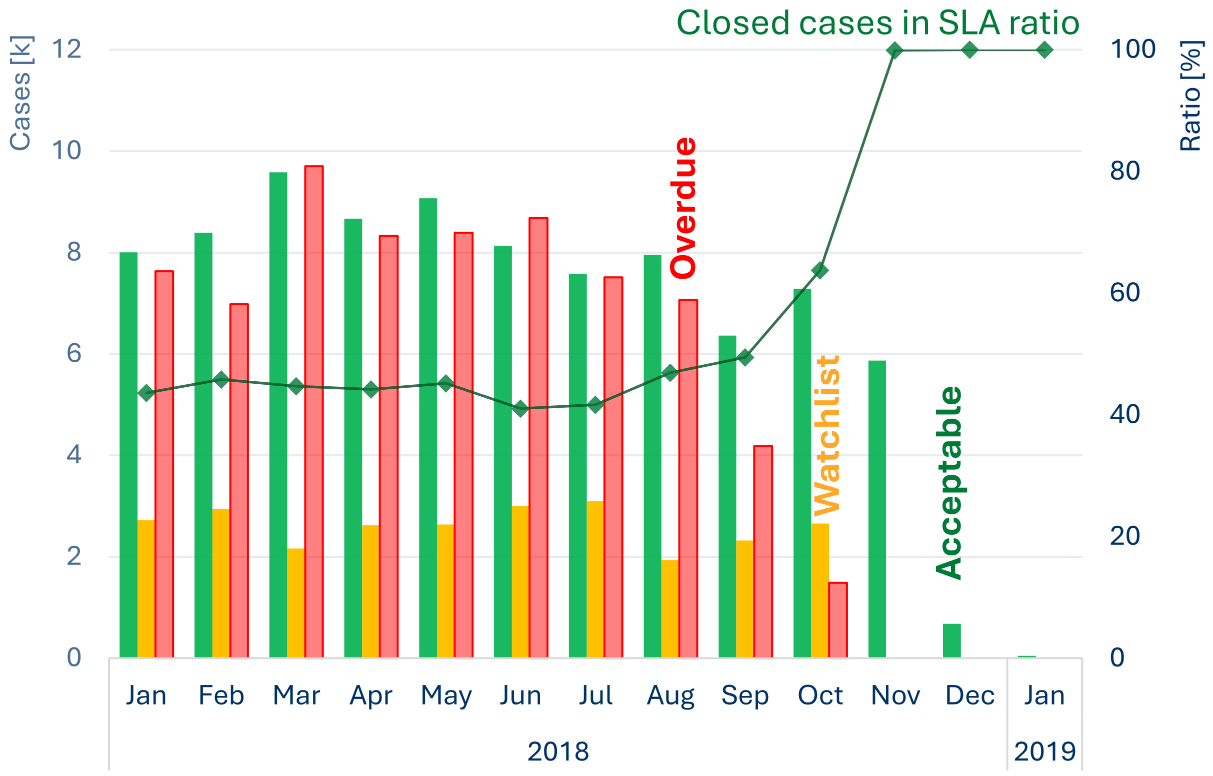
Closed vs Open Cases

The descreasing share of closed cases indicates a growing execution gap despite stable SLA performance.



Closed Cases distribution by category

Historically, **over 80% of cases** are eventually **closed**, **with 40–45% of them meeting the SLA target.**



Although the latest months show a **high share of closures within SLA**, the result is likely affected by immature data.

Is SLA performance deteriorating?

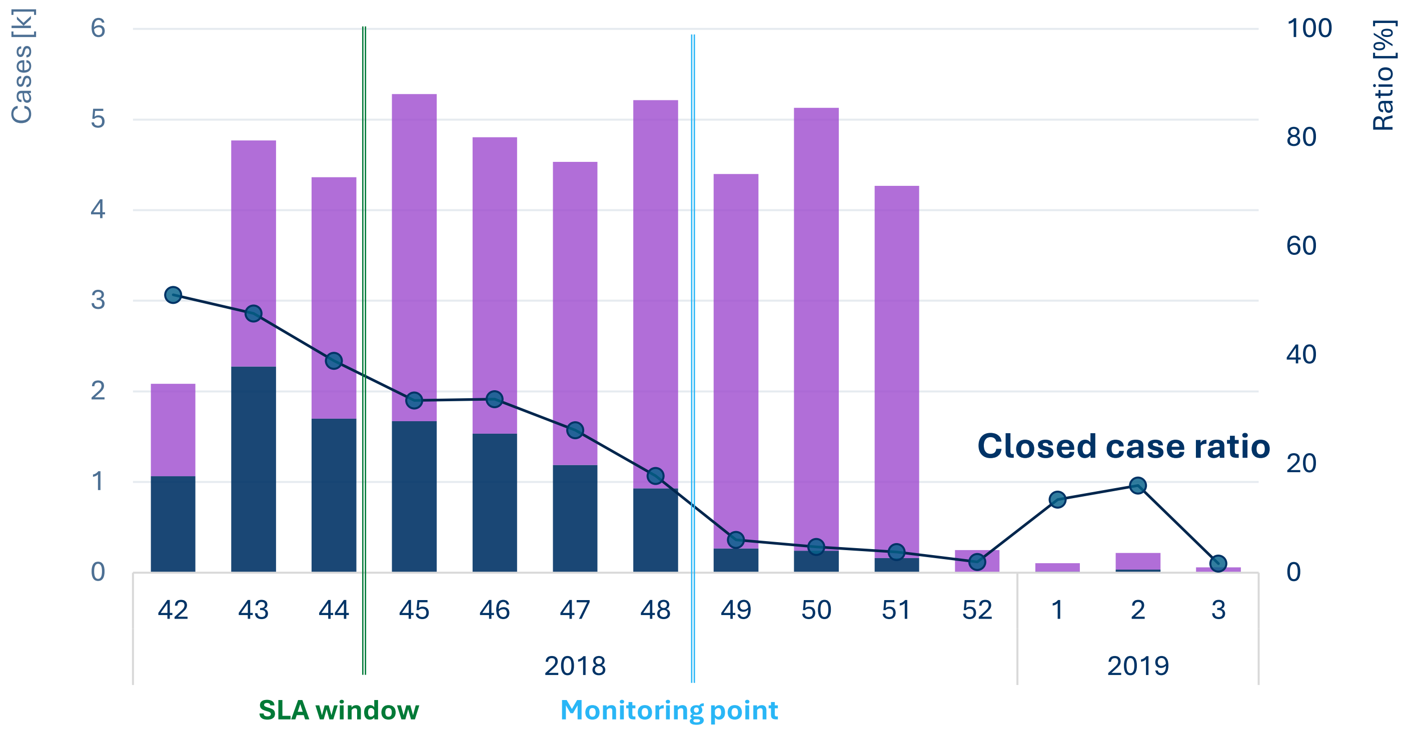
Lead time vs backlog risk

Snapshot status for Jan-2019 + 2H 2018.

Weekly Trend

Closed vs Open Cases

Open cases in the **8–11 week range** should be treated as a priority for monitoring and potential intervention.



Closed cases duration

Closed Cases	Duaration Days	
Stage	Median	P90
<= 75 days	51	70
> 75 days	105	164

Median of days duration for **closed cases** with SLA is **51 days (~7 weeks)**.

Backlog overview

Backlog	Total Number	Ratio [%]
All (>90 days)	19,909	37.07

Backlog constitutes **37% of open cases**.

Backlog Breakdown	Total Number	Ratio [%]
91-120 days	8,374	16
121-150 days	4,111	8
>150 days	7,424	14

Ratio as % of cases within all open cases

There is a significant accumulation of backlogs, especially **in segments >150 days.**

The issue is not SLA capability, but insufficient throughput leading to backlog accumulation.

Where are the bottlenecks in the process?

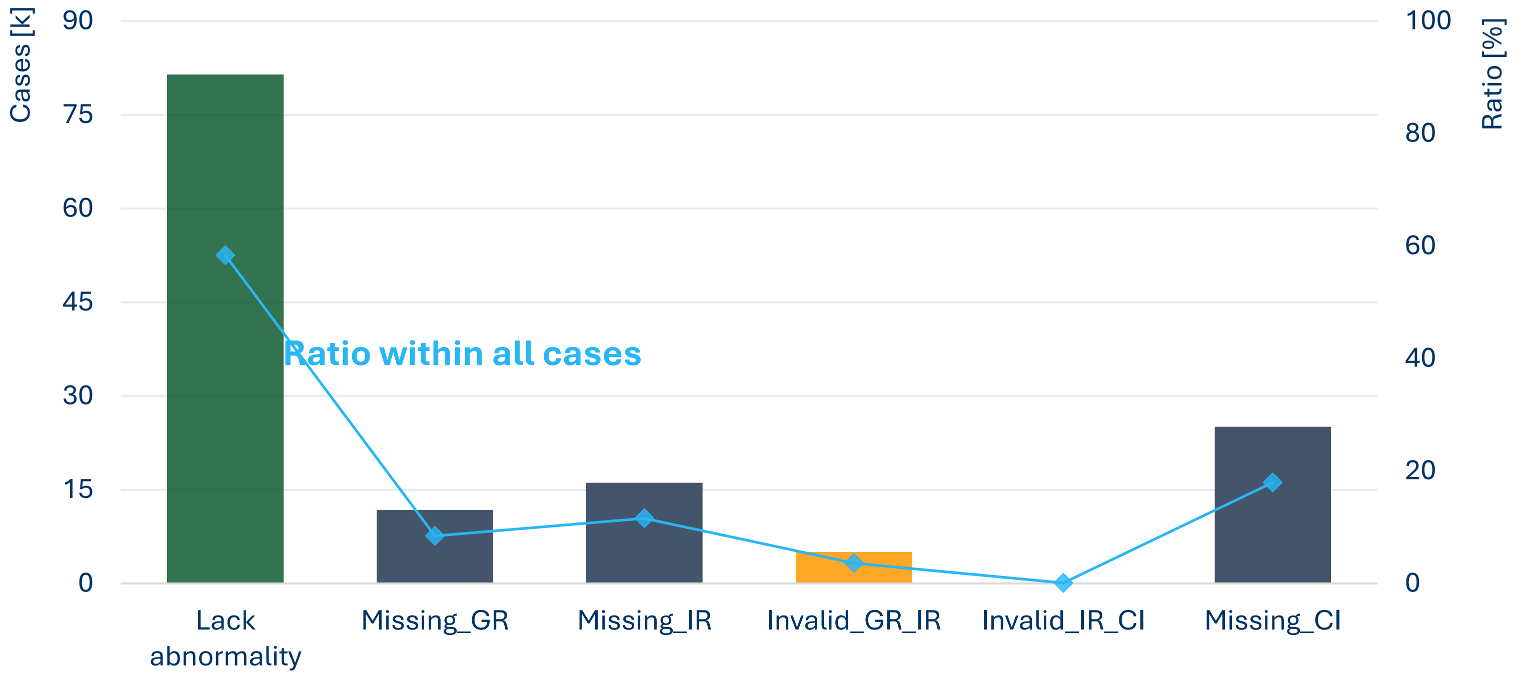
Data validation

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on all cases.

Abnormalities & Missingness distribution

Status for all cases

Within presented cases **almost 60% have all steps and correct data.**
Below 4% cases have **invalid data**. These cases was excluded from next analysis.



Primary_Issue	Cases	Ratio [%]
Lack abnormality	81,438	58.36
Missing_GR	11,766	8.43
Missing_IR	16,130	11.56
Invalid_GR_IR	5,034	3.61
Invalid_IR_CI	123	0.09
Missing_CI	25,058	17.96

- Primary Issue = earliest blocking issue
- Missing = first unavailable milestone
- **Invalid** = negative stage duration

The final stage (IR → CI) shows the highest missingness and should be prioritised for root-cause validation.

Where are the bottlenecks in the process?

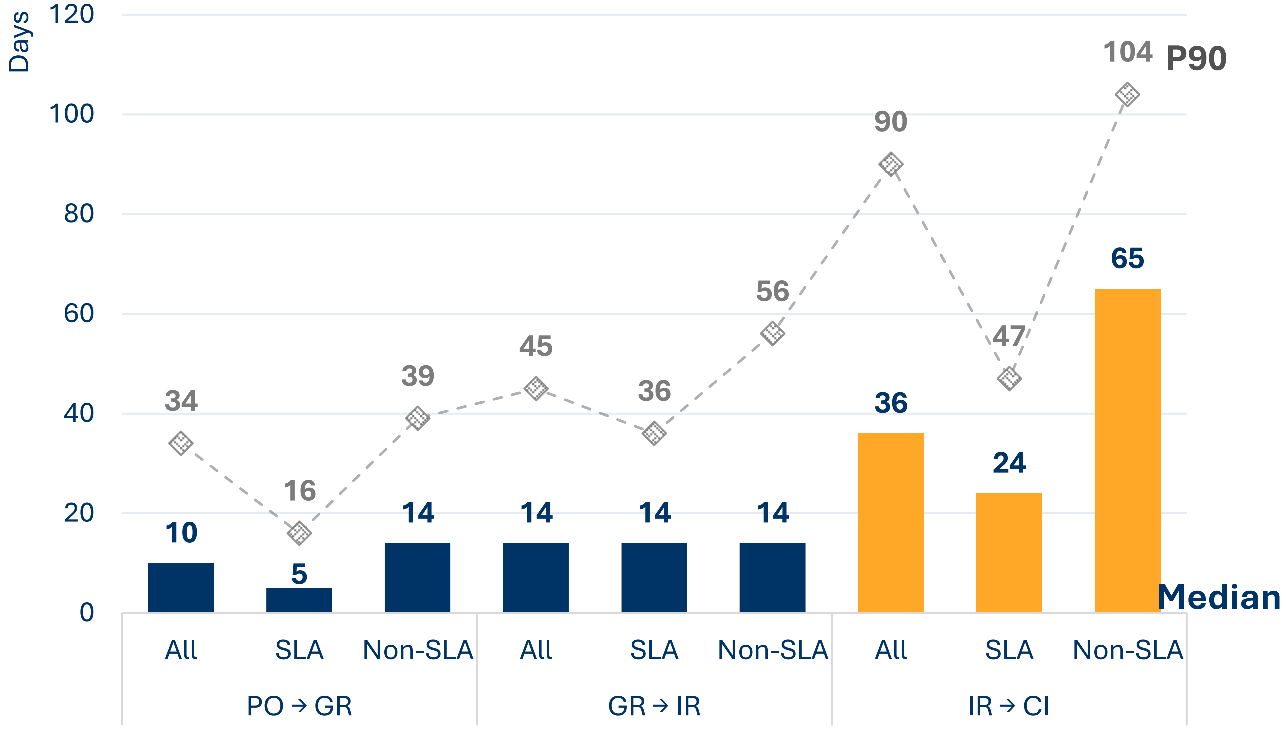
Define stage and impact

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on closed cases only.

Stage duration

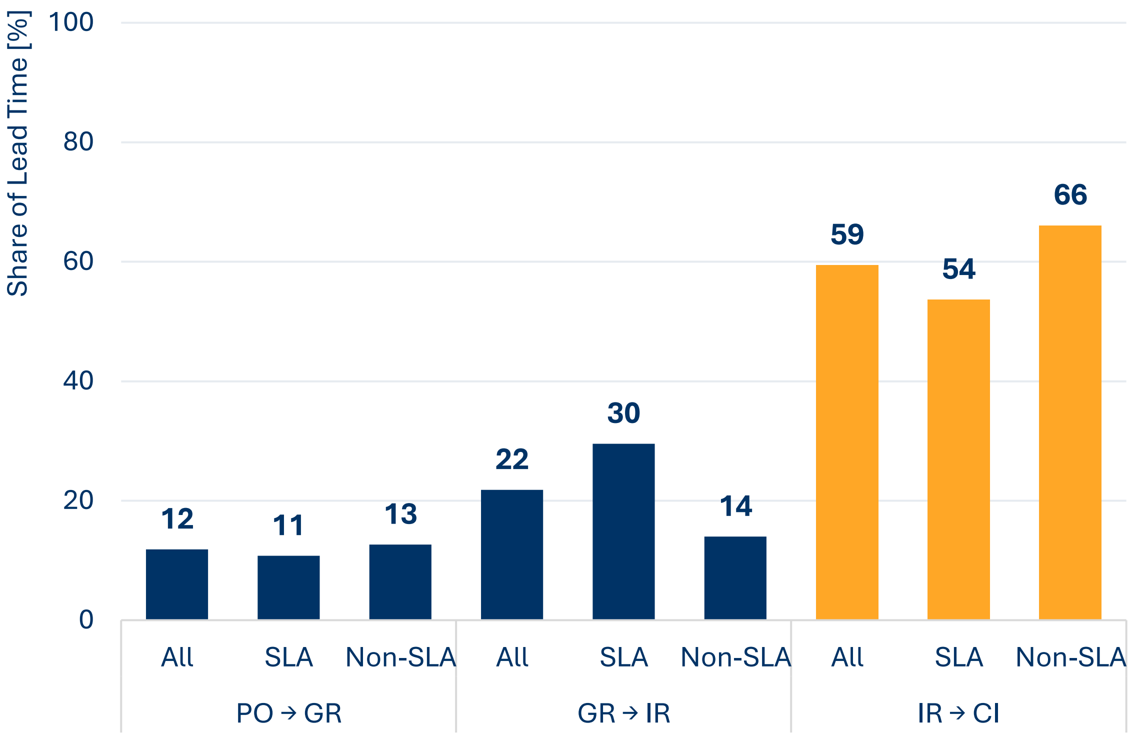
Stage Duration by Process Step

The IR → CI stage has the highest median and P90 duration.



Median Lead Time Contribution by Stage

IR → CI stage contributes more than half of total lead time in delayed cases.



Contribution calculated as stage duration / total case lead time.

The primary bottleneck is the IR → CI stage.

Which segments drive delays?

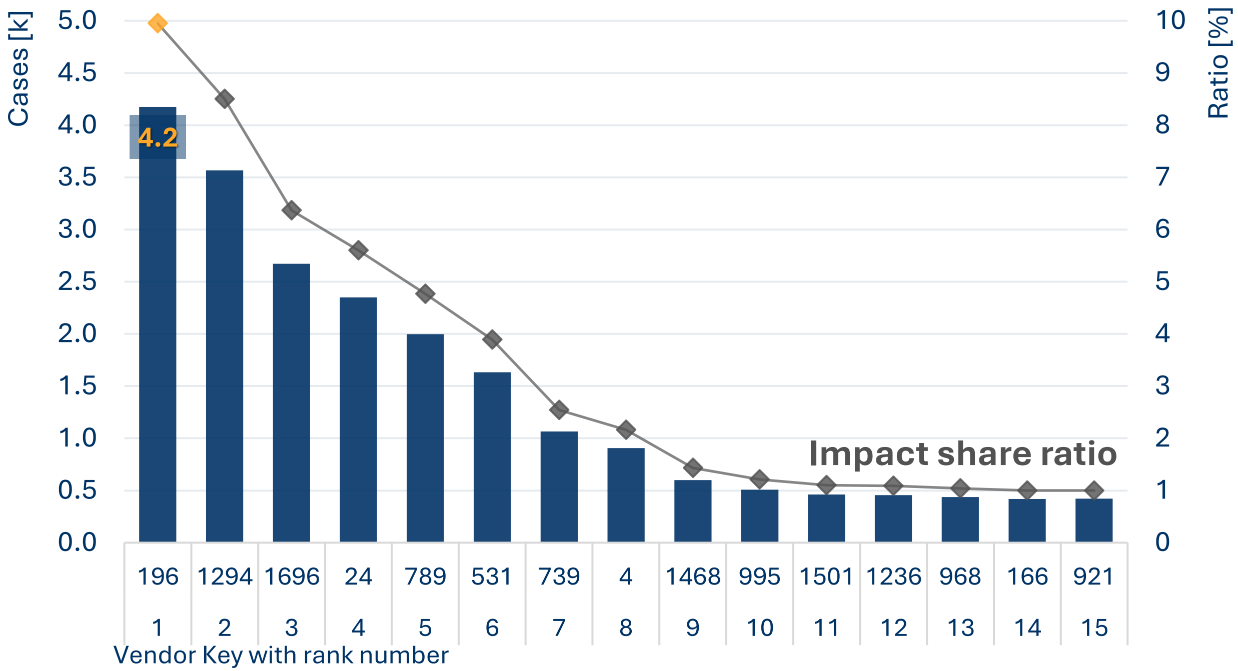
Vendors, companies, spend area

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on closed cases only.

Vendors analysis

Pareto chart – Top 15 Vendors

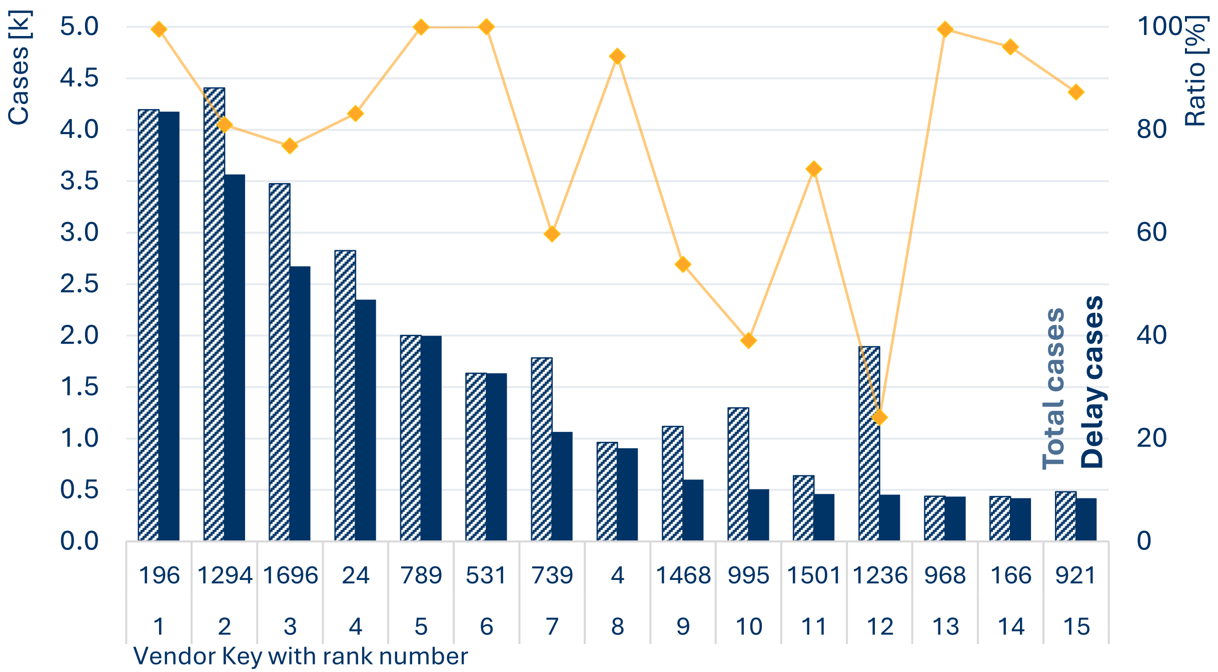
The **largest single vendor** accounts for **10% of all delayed cases** with **almost 4.2 thousand delay cases**.



Impact Share is delayed cases within one vendor / all delayed cases.

Delay Ratio per Vendor

Median delay rate across analysed vendors is **92%**.
134 vendors met the analysis threshold (**> 100 total cases**).



Delay ratio as delayed cases / total cases for one vendor

Delay risk is concentrated: top 15 vendors drive 52% of delayed cases.

Vendor IDs used due to unavailable supplier names in source data.

Which segments drive delays?

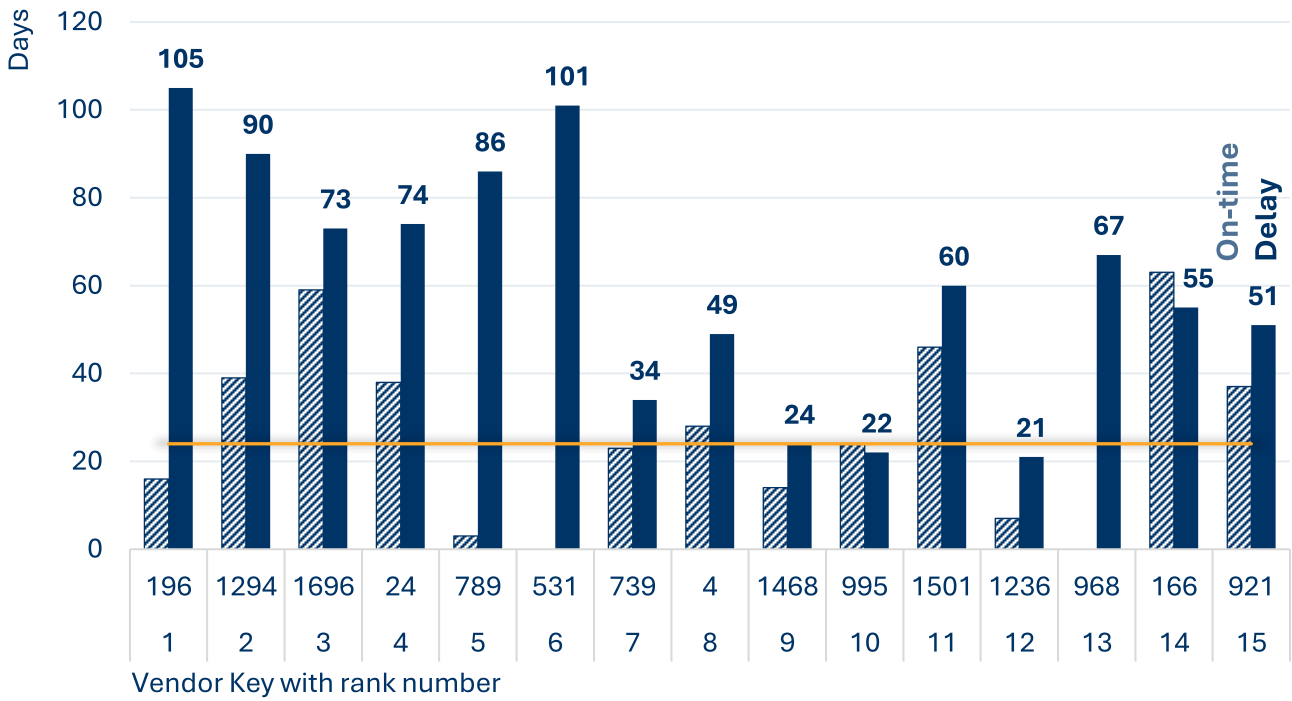
Vendors, companies, spend area

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on closed cases only.

Vendors analysis

IR → CI Median Duration by Vendor (On-time vs Delayed)

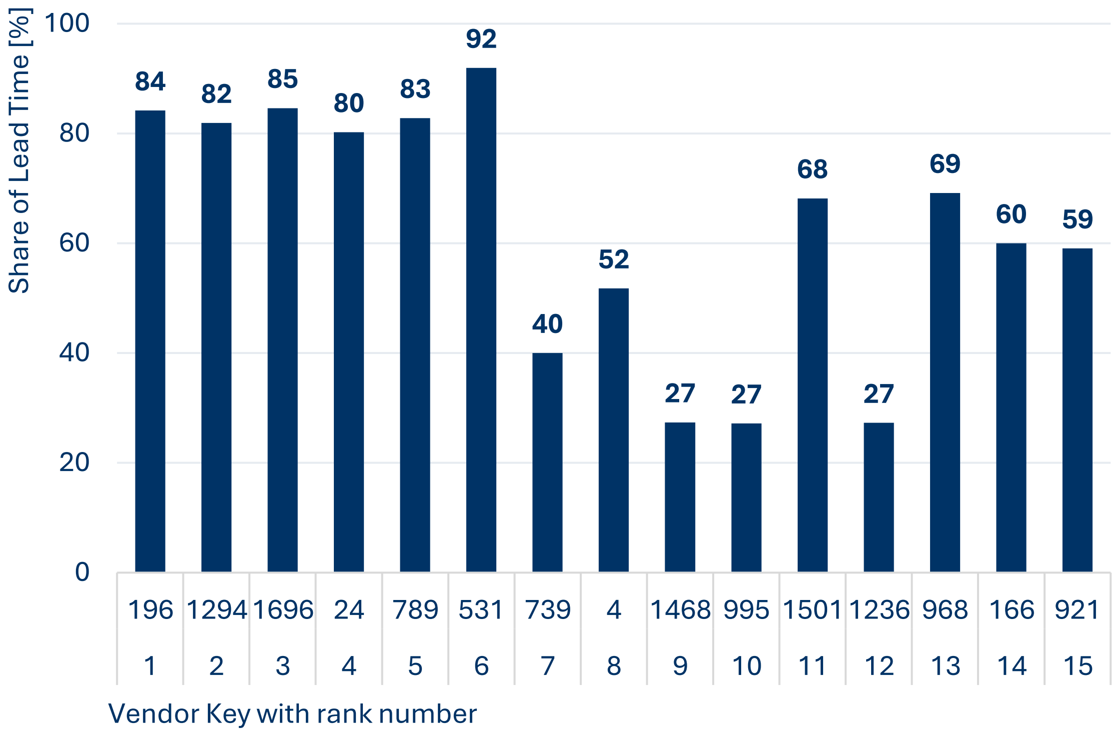
Vendors with delayed cases perform materially worse than **SLA benchmark (24 days)**.



SLA benchmark is median duaration of on-time cases.

IR → CI Contribution to Delay by Vendor

IR → CI remains the main bottleneck across vendors.



Delays are not only supplier-driven; internal downstream processing is also a key constraint.

Vendor IDs used due to unavailable supplier names in source data.

Which segments drive delays?

Vendors, companies, spend area

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on closed cases only.

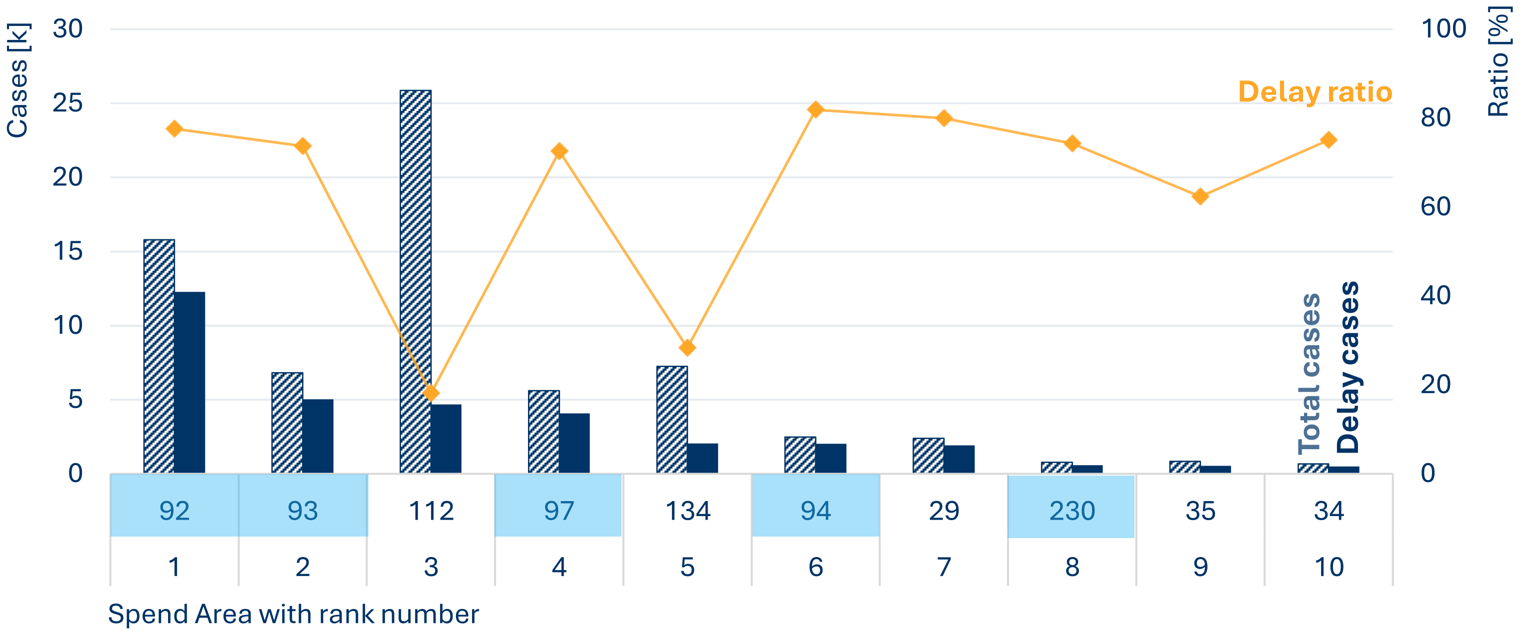
Spend areas analysis

Spend Area with sub category	Impact Share [%]
Packaging	**57.19**
92: Labels	29.25
93: Metal Containers & Lids < 30L	12.00
97: Plastic Containers & Lids < 30L	9.72
94: Packaging - Other	4.84
230: Labels	1.38
Sales	**11.17**
112: Products for Resale	11.17
Trading & End Products	**4.91**
134: Trading products (old structure)	4.91
Additives	**7.01**
29: Extenders	4.57
35: Surfactants	1.26
34: Rheology & Thixotropic Agents	1.18

Impact Share is delayed cases within one spend area / all delayed cases.

Delay Ratio per Spends area

The **Packaging category** drives the highest impact (**57% of delayed cases**), with "92: Labels" representing the largest individual spend area (29% impact share).
54 spend areas met the analysis threshold (**> 100 total cases**).



Delay ratio as delayed cases / total cases from one spend area.

Delay risk is concentrated: top 10 spends areas drive 82% of delayed cases.

Which segments drive delays?

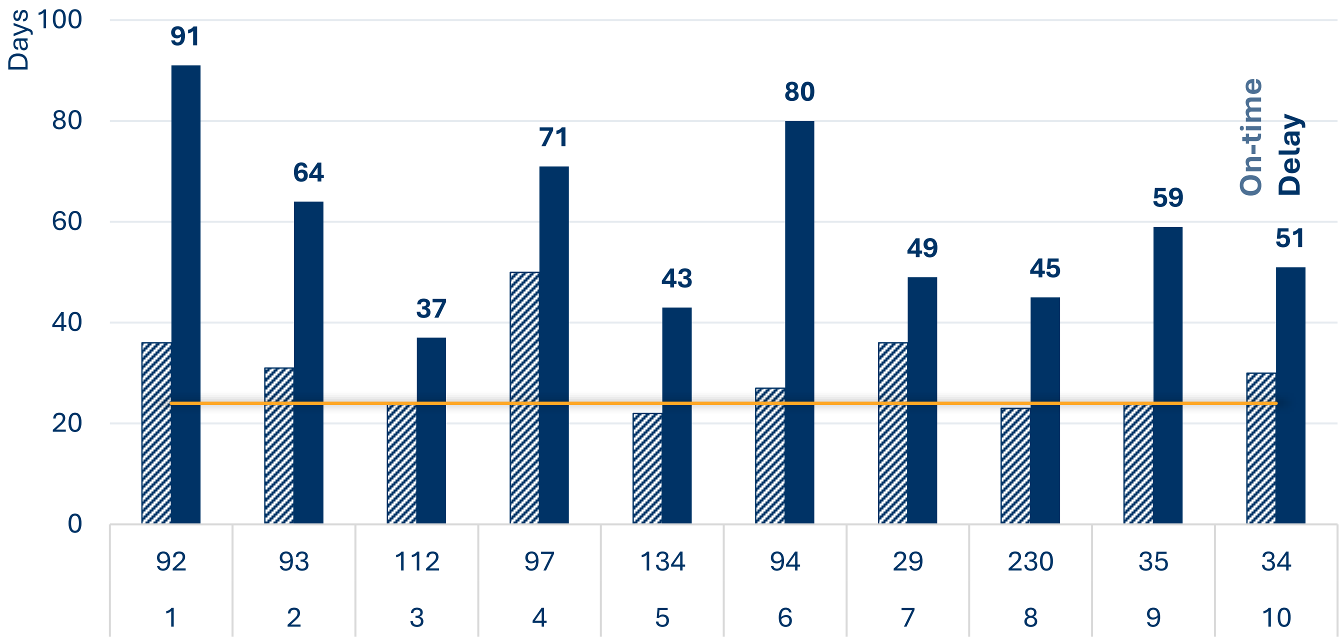
Vendors, companies, spend area

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on closed cases only.

Spend areas analysis

IR → CI Median Duration by Spend area (On-time vs Delayed)

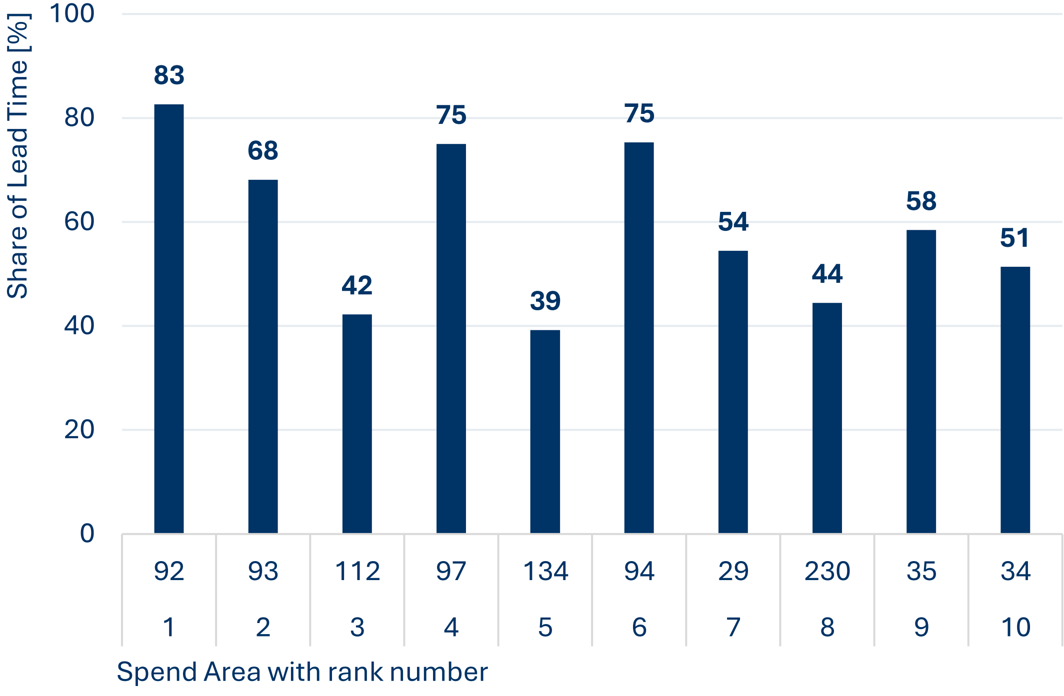
Spend areas with delays have worst results than **SLA benchmark (24 days)**.



SLA benchmark is median duaration of on-time cases.

IR → CI Contribution to Delay by Spend areas

The IR → CI step is still main bottleneck, **median delay rate** across analysed spend areas is **56%.**



The problem in spend area is still for IR → CI.

Which segments drive delays?

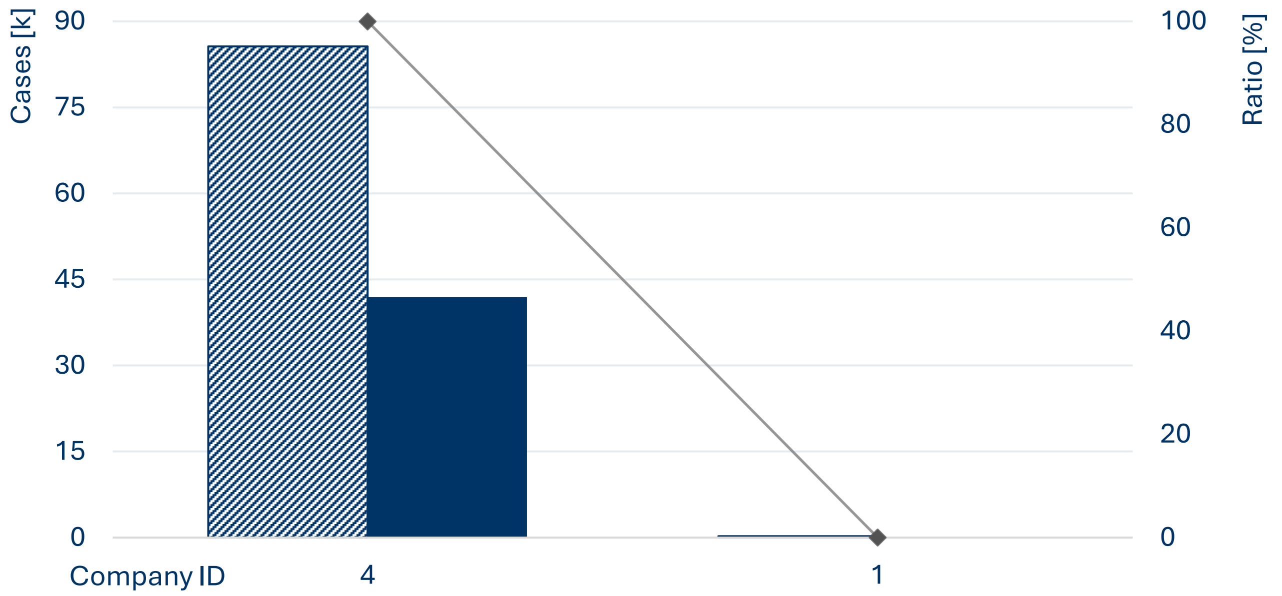
Vendors, companies, spend area

Snapshot status for Jan-2019 + 2H 2018.
Analysis based on closed cases only.

Companies analysis

Companies

Company 4 accounts for **99.99% of delayed cases**.
Delay exposure is highly concentrated in one operating entity.



Impact Share is delayed cases within company / all delayed cases.

Company ID	Total Cases	Delay Cases	Impact Share
4	85,610	41,941	99.99
1	235	4	0.01

Priority review should focus on Company 4 workflow, ownership model, and controls.

Company IDs used due to unavailable supplier names in source data.